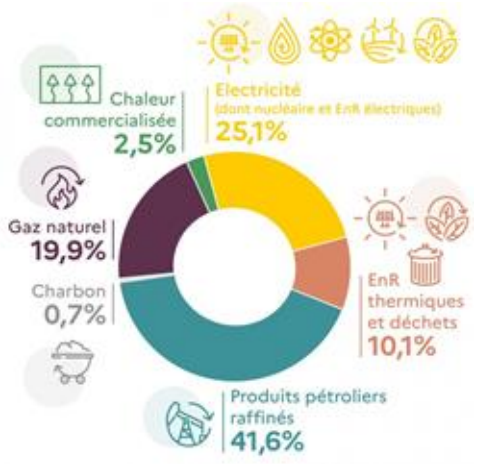
École des Ponts - 1ère année

Bases scientifiques pour la transition énergétique 1

Séance 1 – Énergie, ressources et climat : la nécessité d'une transition

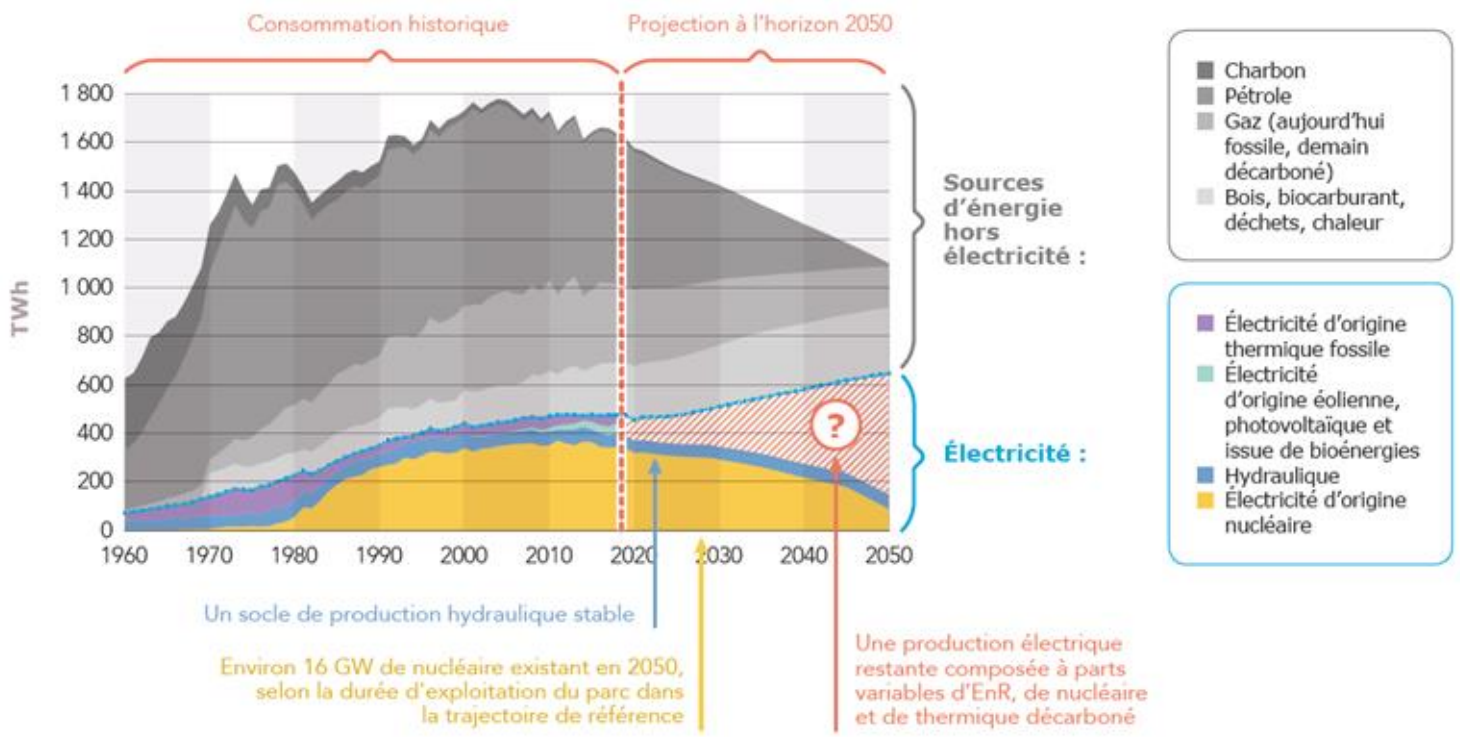


Laurent BROCHARD

11 septembre 2026

École des Ponts ParisTech

Contexte : la transition énergétique



Source : RTE (2022) Futurs énergétiques de la France

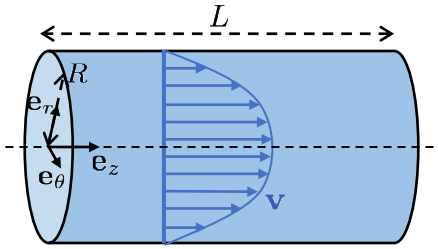
→Socle scientifique indispensable pour appréhender les enjeux de la transition énergétique ?

- problématique technique → sciences physiques : ENRJ1 (sept.-oct.)
- **et** sociétale → sciences humaines : ENRJ2 (fév.-mars)

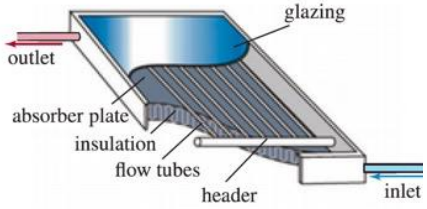
Contenu sciences physiques

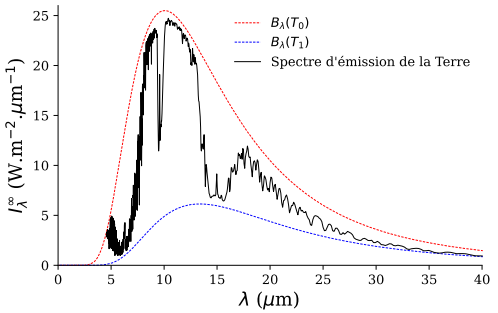
- Séance 1 : généralités & TD préliminaire



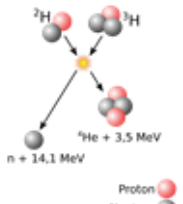


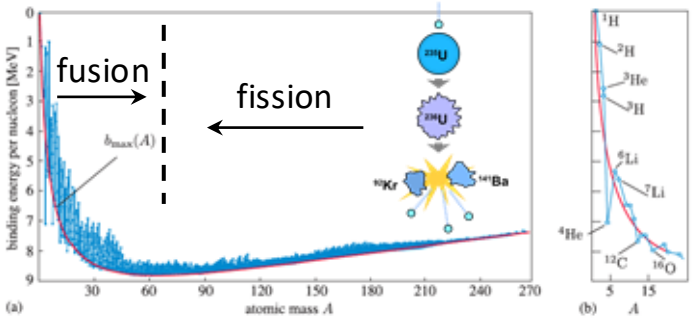
- Séance 2-3 : énergie du rayonnement

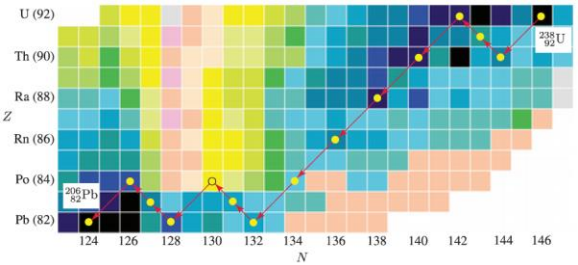


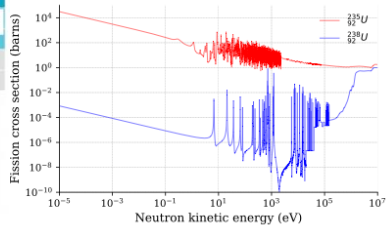


- Séance 4-5 : énergie nucléaire

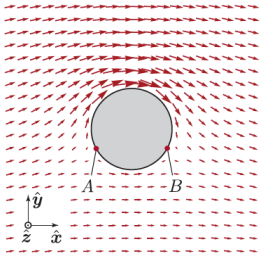


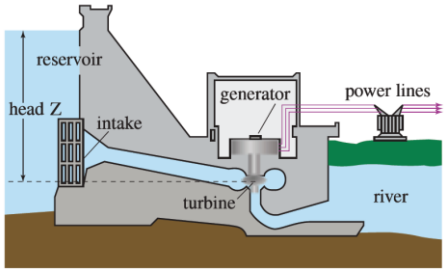


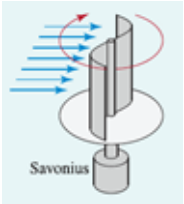




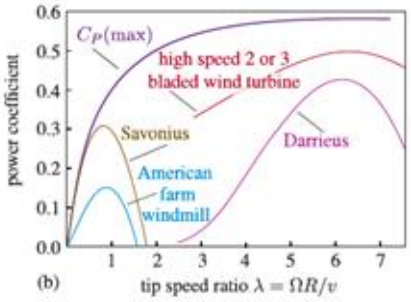
- Séance 6&7 : énergie des fluides











Organisation du module

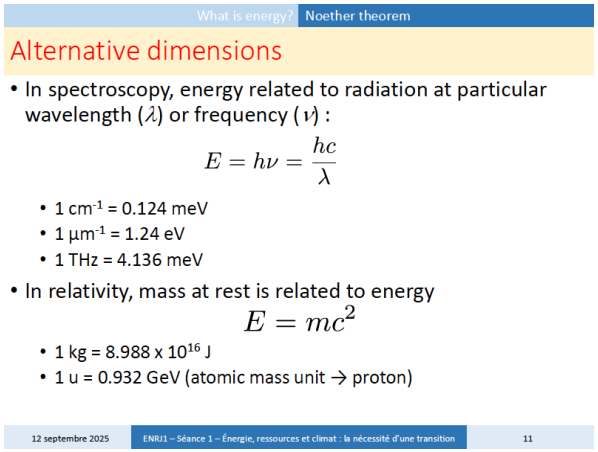
- Format : amphi (S1, S2, S4, S6) + TD (toutes les séances)
- Validation : examen écrit (06/11)
- Tous les documents sur Educnet : https://educnet.enpc.fr/course/view.php?id=1761
- Equipe enseignante :
  - Groupe 1 (B004) - Laurent BROCHARD (Navier, resp. module)
  - Groupe 2 (B005) - Louis CARILLO (CERMICS) & Anaël LEMAITRE (Navier)
  - Groupe 3 (B102) - Christophe CHEVALIER (Univ. Gustave Eiffel)
  - Groupe 4 (B103) - Nicolas DAUNAIS (Navier)
  - Groupe 5 (V002) - Caroline KOPECZ-MULLER (Navier)
  - Groupe 6 (V003) - Mathias LEBIHAIN (Navier)
  - Groupe 7 (P202) - Bilal MAHMOUD HAWCHAR (Navier)
  - Groupe 8 (P203) - Rahima SIDI-BOULENOUAR (Navier)
  - Daniel SUCHET (Polytechnique) – séance 2

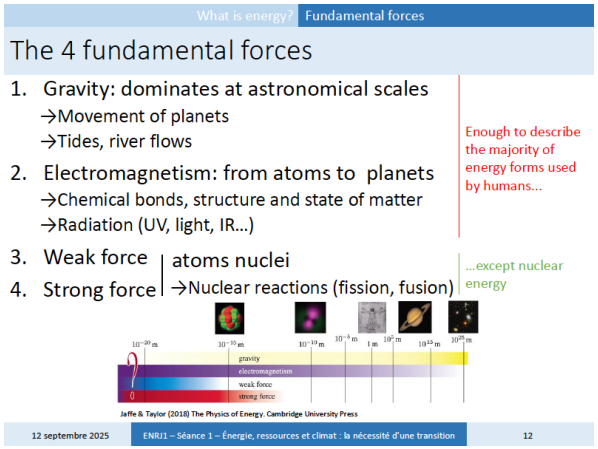
Planning du module

- ***11/09 – Énergie, ressources et climat : la nécessité d'une transition***
- 18/09 – Rayonnement 1 (Amphi + TD)
- 25/09 – Rayonnement 2 (TD seulement)
- 02/10 – Énergie nucléaire 1 (Amphi + TD)
- 09/10 – Énergie nucléaire 2 (TD seulement)
- 16/10 (matin) – Énergie du mouvement des fluides 1 (Amphi + TD)
- 16/10 (après midi) – Énergie du mouvement des fluides 2 (TD seulement)
- 06/11 – Examen

Remarque préliminaire sur le contenu des cours

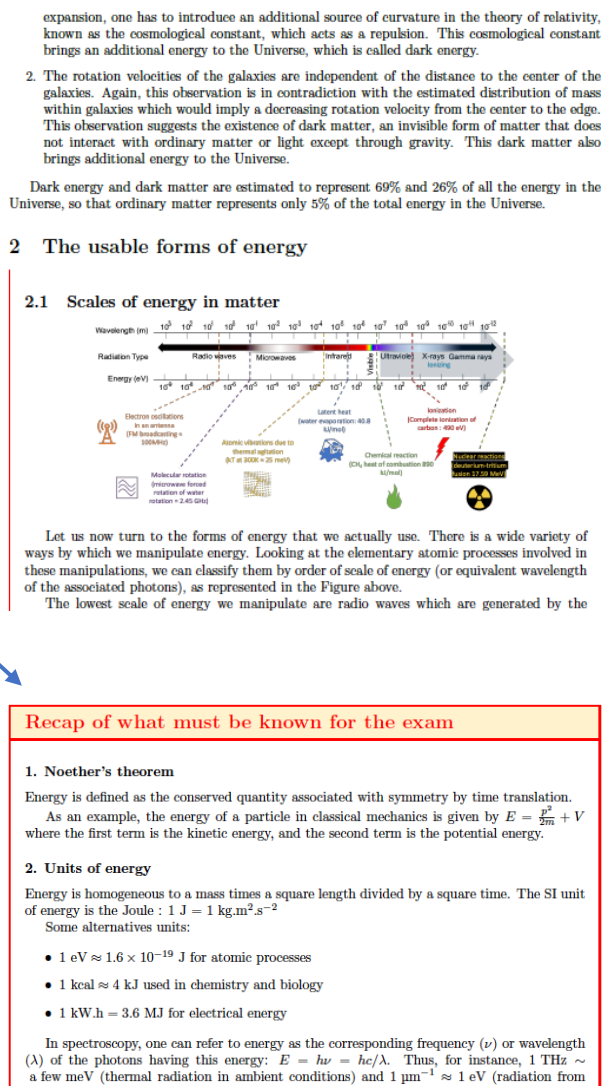
- Les amphis et le poly contiennent plus que ce qu'il est nécessaire de connaitre pour l'examen
- Les TD doivent être maitrisés et sont représentatifs de ce qui peut être proposé à l'examen
- Code couleur slide
  - Doit être connu
  - Non nécessaire





Remarque préliminaire sur le contenu des cours

- Code couleur poly
  - Doit être connu
  - Non nécessaire
- Récapitulatifs des points importants en fin de chapitre
- Quelques précisions
  - Les valeurs numériques précises des constantes fondamentales n'ont pas besoin d'être connues par cœur
  - Mais les ordres de grandeur doivent être connus



expansion, one has to introduce an additional source of curvature in the theory of relativity, known as the cosmological constant, which acts as a repulsion. This cosmological constant brings an additional energy to the Universe, which is called dark energy.

2. The rotation velocities of the galaxies are independent of the distance to the center of the galaxies. Again, this observation is in contradiction with the estimated distribution of mass within galaxies which would imply a decreasing rotation velocity from the center to the edge. This observation suggests the existence of dark matter, an invisible form of matter that does not interact with ordinary matter or light except through gravity. This dark matter also brings additional energy to the Universe.

Dark energy and dark matter are estimated to represent 69% and 26% of all the energy in the Universe, so that ordinary matter represents only 5% of the total energy in the Universe.

2 The usable forms of energy

2.1 Scales of energy in matter

Let us now turn to the forms of energy that we actually use. There is a wide variety of ways by which we manipulate energy. Looking at the elementary atomic processes involved in these manipulations, we can classify them by order of scale of energy (or equivalent wavelength of the associated photons), as represented in the Figure above.

The lowest scale of energy we manipulate are radio waves which are generated by the

Recap of what must be known for the exam

1. Noether's theorem

Energy is defined as the conserved quantity associated with symmetry by time translation.

As an example, the energy of a particle in classical mechanics is given by $E = \frac{p^2}{2m} + V$ where the first term is the kinetic energy, and the second term is the potential energy.

2. Units of energy

Energy is homogeneous to a mass times a square length divided by a square time. The SI unit of energy is the Joule : $1\ \mathrm{J} = 1\ \mathrm{kg.m^2.s^{-2}}$

Some alternatives units:

- $1\ \mathrm{eV} \approx 1.6 \times 10^{-19}$ J for atomic processes
- 1 kcal ≈ 4 kJ used in chemistry and biology
- 1 kW.h = 3.6 MJ for electrical energy

In spectroscopy, one can refer to energy as the corresponding frequency (ν) or wavelength (λ) of the photons having this energy: $E = h\nu = hc/\lambda$. Thus, for instance, 1 THz ∼ a few meV (thermal radiation in ambient conditions) and $1\ \mu\mathrm{m}^{-1} \approx 1$ eV (radiation from

Plan of the lesson

1. What is energy?

2. The usable forms of energy

3. Exploitation of energy

Plan of the lesson

1. What is energy?
   - Noether theorem
   - Fundamental forces
   - Energy-mass, dark energy, dark matter
2. The usable forms of energy
3. Exploitation of energy

Conserved quantity

- Noether: To every continuous symmetry of the action of a physical system corresponds a conserved quantity
  - Symmetry by spatial translation: conservation of linear momentum
  - Symmetry by spatial rotation: conservation of angular momentum
  - Symmetry by time translation: conservation of energy
- *Any isolated physical system with time-invariant quantum or classical physical laws has a conserved quantity, which is called energy.*

Quantum mechanics

$$\widehat{H} = -\frac{\hbar^2}{2m}\Delta + V$$

kinetic potential

Classical mechanics

$$U = \frac{p^2}{2m} + V$$

kinetic potential

- Potential? → fundamental forces

Units

SI unit : 1 J = 1 $kg.m^2.s^{-2}$

- For atomic scale : 1 eV = 1.60218 x 10^{-19} J

  Energy change of an electron moved across an electric potential of 1V

- Biology / chemistry : 1 cal = 4.184 J or 1 kcal = 1 Cal = 4 184 J

  Heat to raise the temperature 1 g of liquid water by 1°C

- Imperial : 1 Btu (British thermal unit) = 1 055 J

  Heat to raise the temperature of one pound of water by one degree Fahrenheit

- Electricity : 1 kW.h = 3.6 MJ

  Energy corresponding to a power of 1kW consumed/produced during one hour

- Miscellaneous

  1 toe (ton oil eq.) = 7.33 boe (barrel oil eq.) = 41.868 GJ – *Energy released by burning 1 ton of crude oil*

  1 ton of TNT = 10^9 cal = 4.184 GJ – *Energy released upon explosion of 1 ton of TNT*

  1 Ha (Hartree) = 27.21 eV = 4.36 × 10^{-18} J - *Energy of the hydrogen atom in its ground state (2 x 13.6 eV)*

Alternative dimensions

- In spectroscopy, energy related to radiation at particular wavelength (λ) or frequency (ν) :

$$E = h\nu = \frac{hc}{\lambda}$$

  - 1 cm^{-1} = 0.124 meV
  - 1 μm^{-1} = 1.24 eV
  - 1 THz = 4.136 meV

- In relativity, mass at rest is related to energy

$$E = mc^2$$

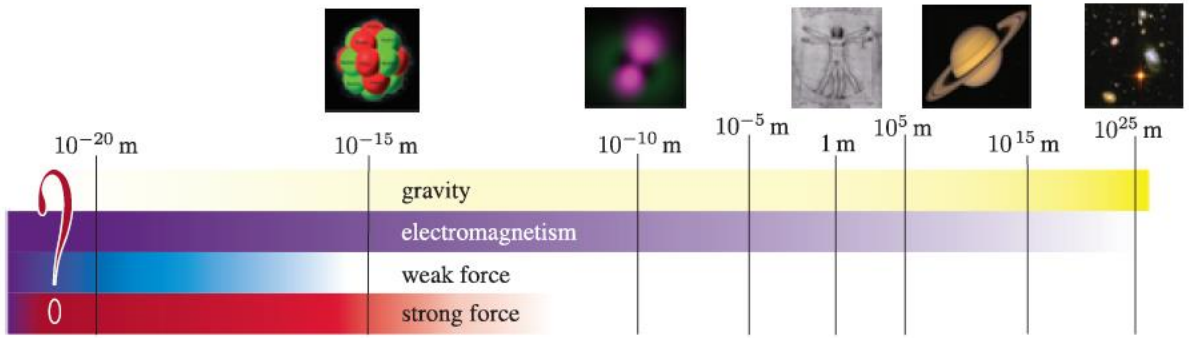
  - 1 kg = 8.988 x 10^{16} J
  - 1 u = 0.932 GeV (atomic mass unit → proton)

The 4 fundamental forces

1. Gravity: dominates at astronomical scales
   → Movement of planets
   → Tides, river flows
2. Electromagnetism: from atoms to planets
   → Chemical bonds, structure and state of matter
   → Radiation (UV, light, IR…)
3. Weak force | atoms nuclei
4. Strong force | → Nuclear reactions (fission, fusion)

Enough to describe the majority of energy forms used by humans…

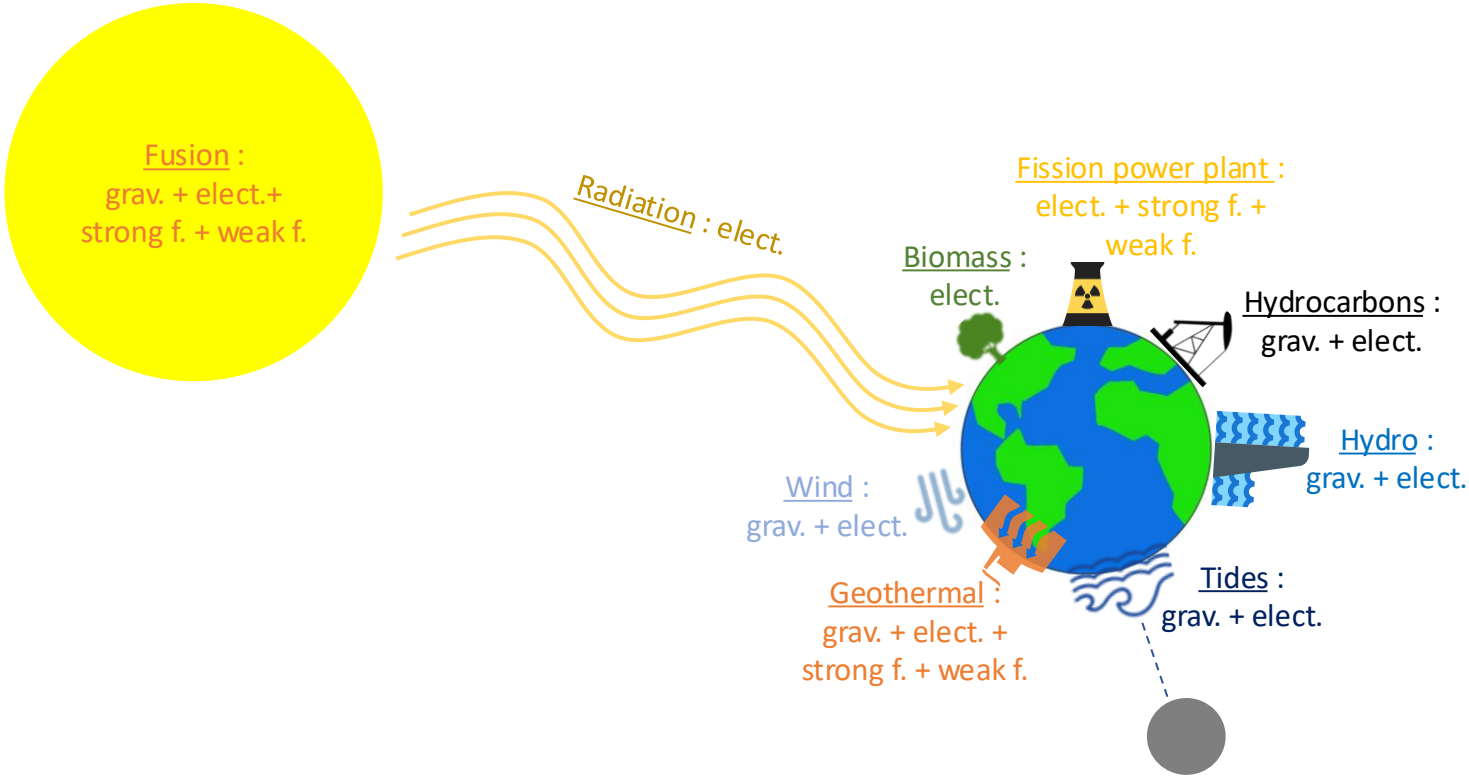
…except nuclear energy



Jaffe & Taylor (2018) The Physics of Energy. Cambridge University Press

Conjunction between the forces

- Majority of processes = conjunction between the forces
- Sun is the primary source for all forms of energy used except geothermal, tidal and nuclear energies.



Strength and range

Gravity	Electromagnetism	Strong force	Weak force
$E_{grav} = -G\frac{m_1 m_2}{r}$	$E_{elect} = \frac{1}{4\pi\epsilon_0}\frac{q_1 q_2}{r}$	$E_{strong/weak} = -\frac{1}{4\pi\epsilon_0}\frac{(g_{strong/weak})^2}{r}e^{-r/b_{strong/weak}}$ "Yukawa potential"	
•between all particles •origin: space-time curvature (relativity)	•between charged particles •quantified (< Å)	•between quarks/hadrons (e.g. protons neutrons)	•between fermions (and also Higgs boson and neutrinos)
•G = 7.10^{-11} J.m^{-1}.kg^{-2} → weak	• $(4\pi\epsilon_0)^{-1}$ = 9.10^{9} J.m^{-1}.C^{-2} → strong	•$g_{strong}^2 \approx 10\ e^2$ → stronger than elect	•$g_{weak}^2 \approx e^2$ → as strong as elect
•$m_1 m_2 > 0$ → universally attractive → continue to act at large distances	•$q_1 q_2$ positive or negative → screening → vanishes at large distances	•$b_{strong} \approx 10^{-15}$ m (1 fm) → extremely short range → size of a nucleus	•$b_{weak} \approx 0.001$ fm → even shorter range

- $E_{grav} \approx 10^{-20}\ E_{elect}$ between macroscopic objects (m = 1 kg, q = 1 C, r = 1 m)
- E_{strong} attraction exceeds E_{elect} repulsion between protons in nuclei.
- Different strong interactions: within protons/neutrons, and between protons/neutrons
- E_{weak} of very small magnitude but explains the nuclear decay, and participates in nuclear fusion/fission

Relativity and mass-energy equivalence

- Special relativity
  - energy-momentum relation $E^2 = c^2\mathbf{p}^2 + m^2c^4$

  →At rest, $E = mc^2$: energy and mass are equivalent

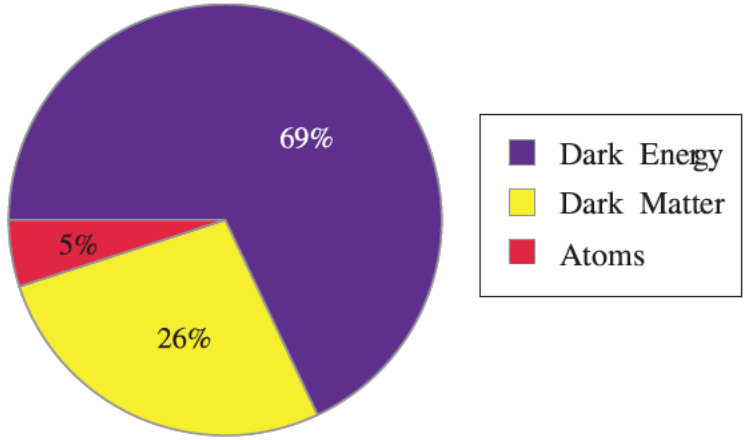
  →In nuclear processes, binding energy in nuclei is often expressed in terms of mass excess (w.r.t isolated nucleons)
- General relativity
  - The accelerating expansion of the universe requires the introduction of a cosmological constant as an additional source of curvature (constant repulsion)

  →Dark energy
- Missing mass
  - Galaxy rotation velocities do not decrease with distance from the center.

  →Dark matter

  (does not interact with ordinary matter and light except through gravity)

Energy in the Universe



Jaffe & Taylor (2018) The Physics of Energy. Cambridge University Press

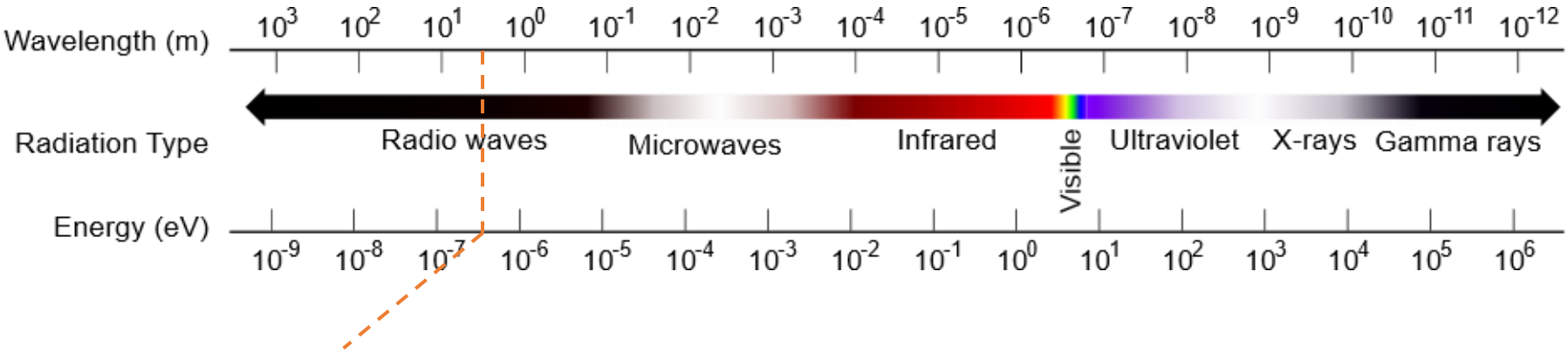
Plan of the lesson

1. What is energy?

2. The usable forms of energy
   - Scales of energy in matter
   - Classification
   - Some orders of magnitude

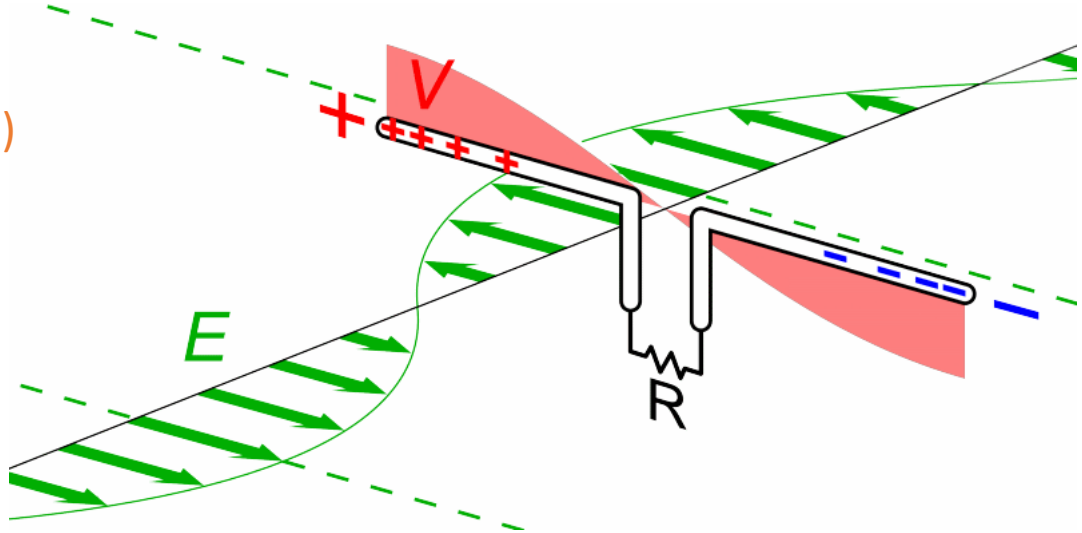
3. Exploitation of energy

By order of increasing energy



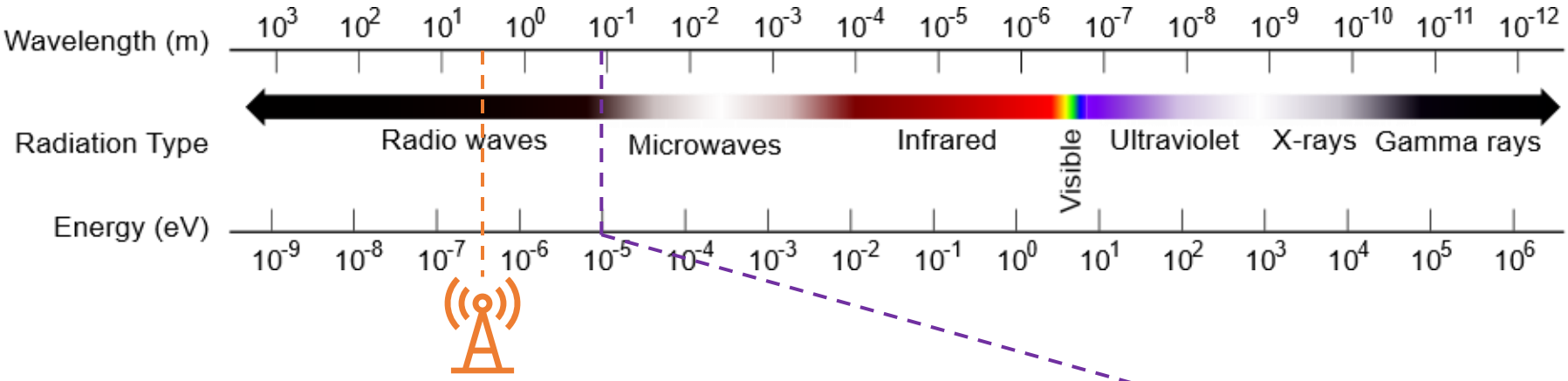
Electron oscillations
in an antenna
(FM broadcasting ≈ 100MHz)

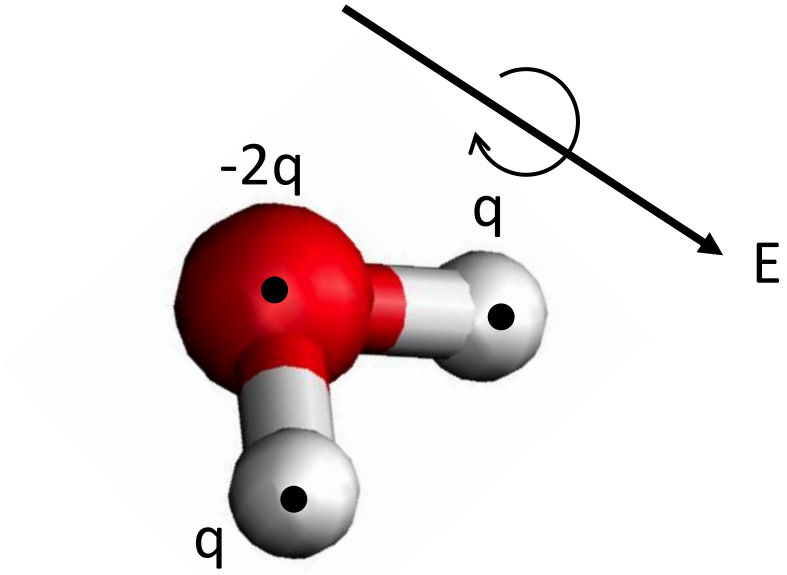




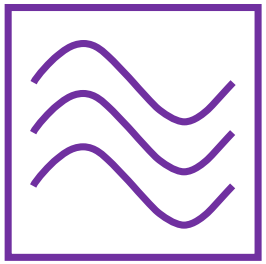
from wikimedia commons (by Chetvorno, CC)

By order of increasing energy

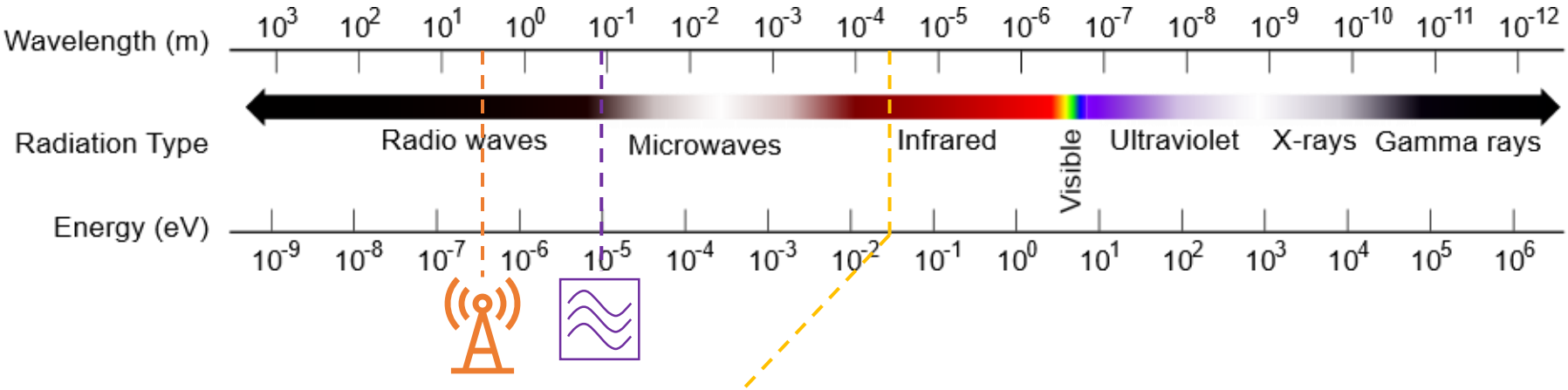




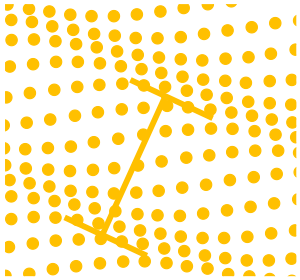
Molecular rotation
(microwave forced rotation
of water rotation = 2.45 GHz)

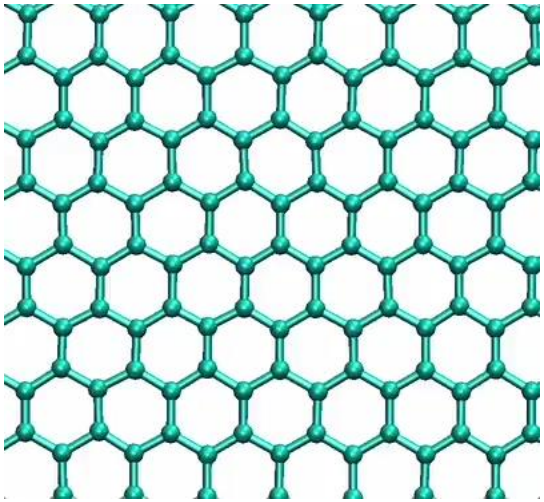


By order of increasing energy

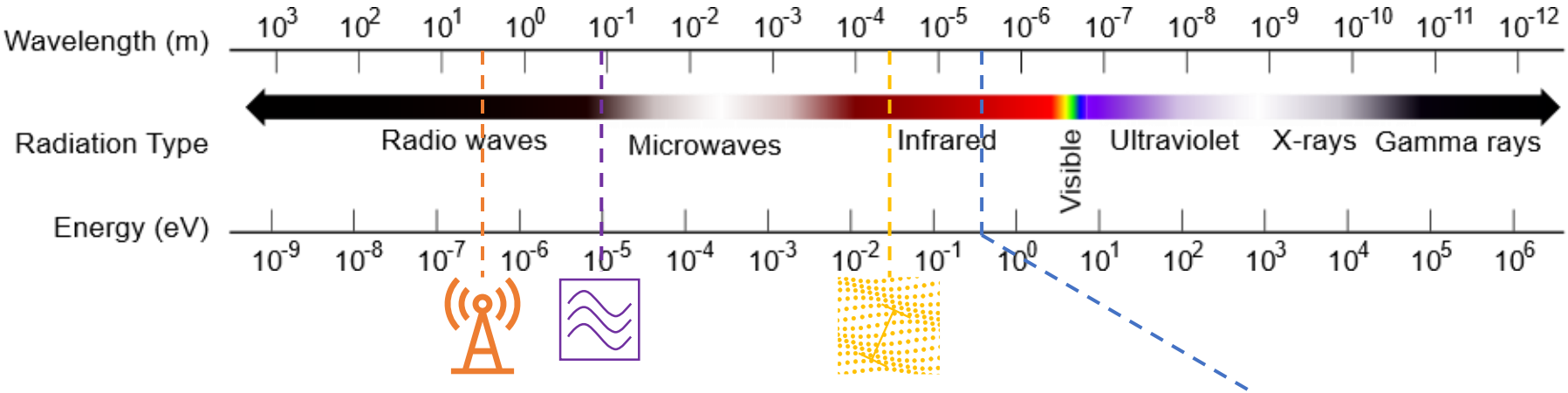


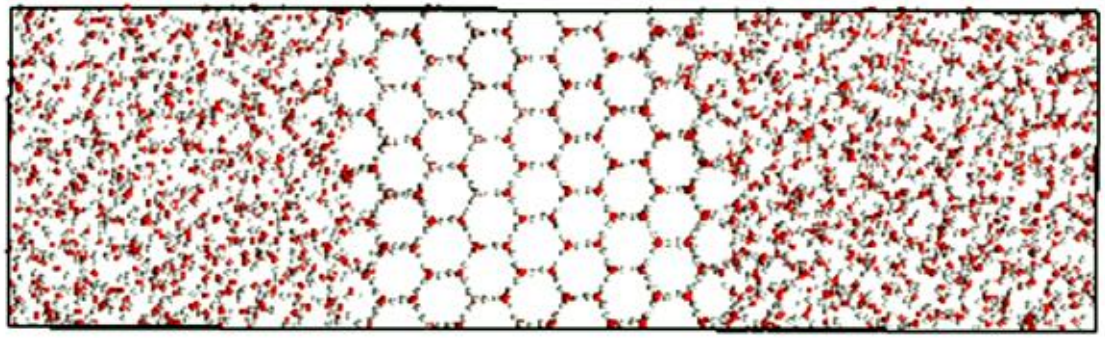
Atomic vibrations due to thermal agitation
(kT at 300K ≈ 25 meV)





By order of increasing energy

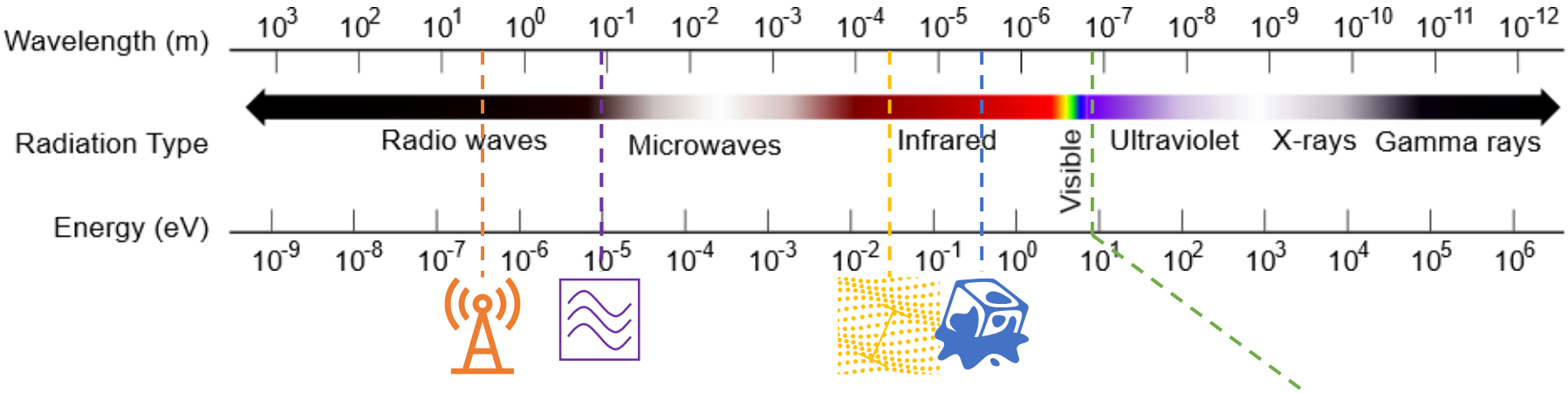


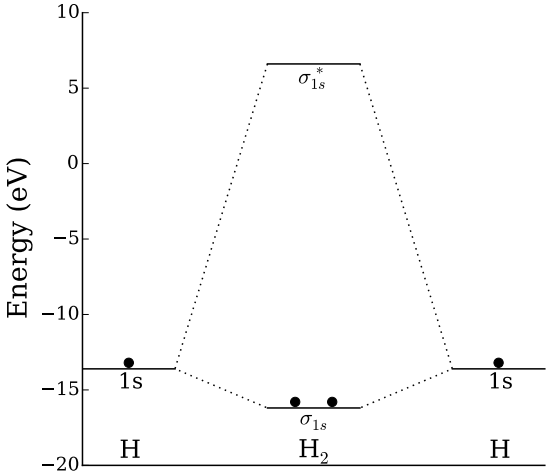


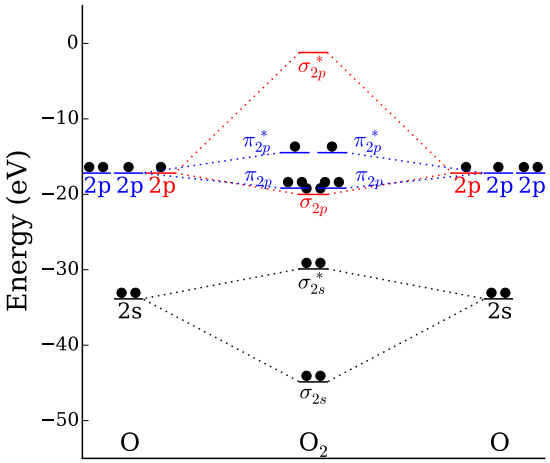
Latent heat
(water evaporation:
40.8 kJ/mol)



By order of increasing energy



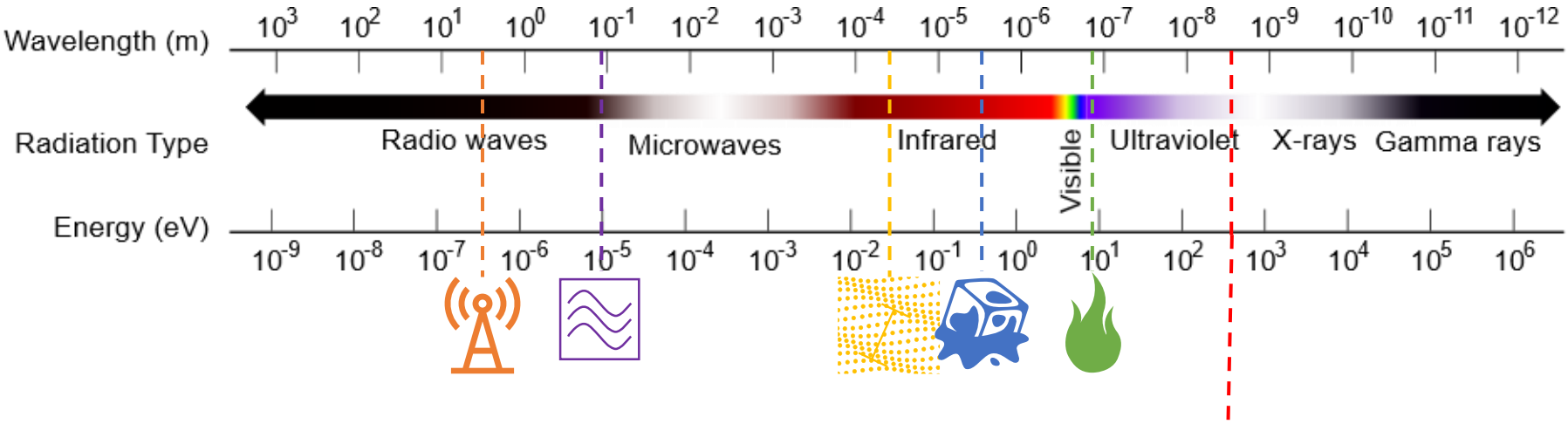




Chemical reaction
(CH_4 heat of combustion
890 kJ/mol)



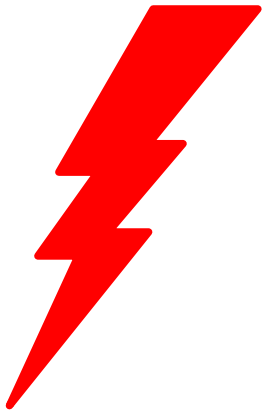
By order of increasing energy



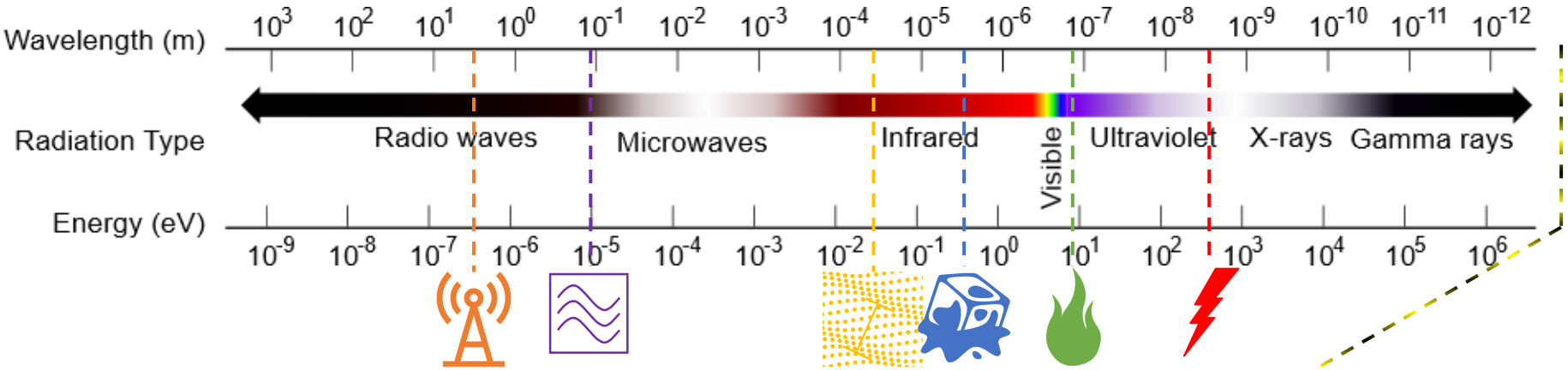


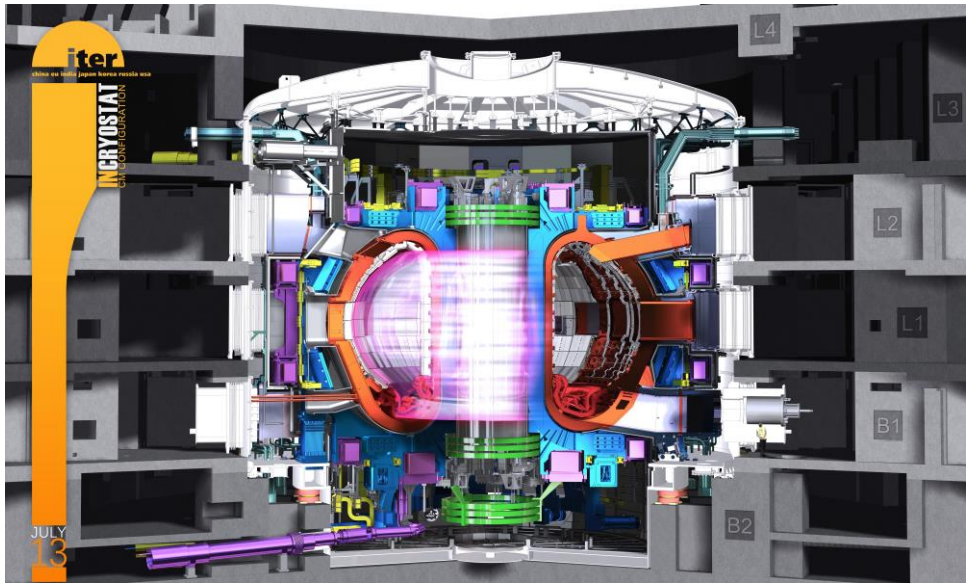
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Ionization
(Complete ionization of carbon : 490 eV)



By order of increasing energy



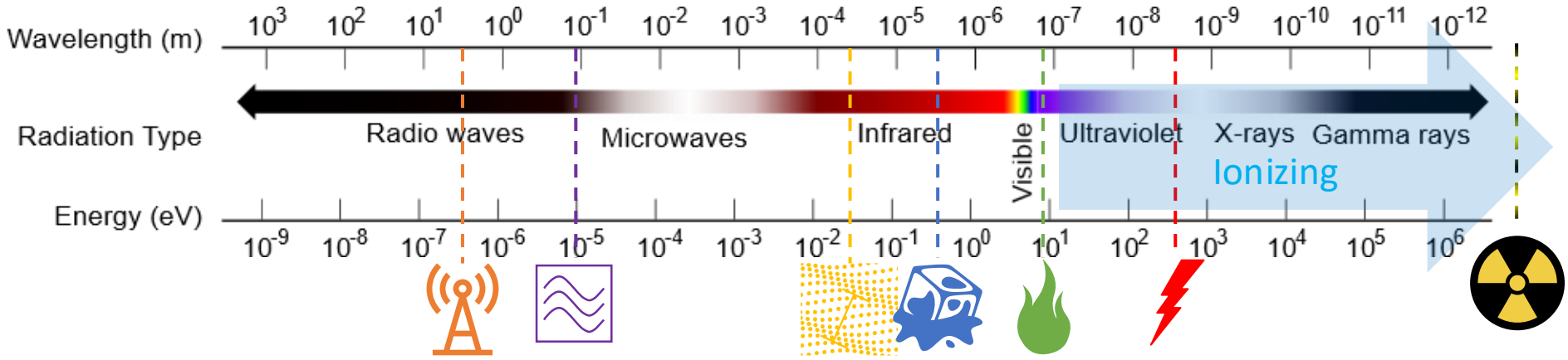


https://www.iter.org/

Nuclear reactions (deuterium-tritium fusion 17.59 MeV)



By order of increasing energy



- 14 orders of magnitudes !
- Ionizing → harmful for life
- Ultimate energy scale : mass
  - annihilation of matter with anti-matter
  - 1 proton : 1 GeV = 10^{9} eV

Forms of energy

Kinetic energy	Potential energy
Motion (ordered kinetic energy) — **Mechanical**	Work (inter-molecular interactions) — **Mechanical**
Thermal (disordered kinetic energy)	
Radiation (photon) — **Electro-magnetic**	Electrical (charged particles) — **Electro-magnetic**
	Chemical (intra-molecular interactions between electrons and nuclei)
	Nuclear (interactions between nucleons)

- Related to fields of applications
- Sometimes ambiguous (e.g., electrical vs work)
- Couplings & transformations

Forms of energy

Kinetic energy	Potential energy
Mechanical: Motion (ordered kinetic energy)	**Mechanical**: Work (inter-molecular interactions)
Thermal (disordered kinetic energy)	
Electro-magnetic: Radiation (photon); Electrical (charged particles)	

- Covered in classe prépa
- Lesson 2 & 3 - 18/09 & 25/09
- Lesson 4 & 5 - 02/10 & 09/10
- Covered in classe prépa
- Lesson 6 & 7 - 16/10
- Future lesson ?

Chemical (intra-molecular interactions between electrons and nuclei)

Nuclear (interactions between nucleons)

Everyday life

- 1 J : a tennis ball falling from 2 m
- 10 J : heat 1g liquid water by 2°C
- 100 J : heat produced by a person at rest during 1 s
- 1 kJ : energy released by a candle burning for 10 s
- 10 kJ : 4 h of electricity consumption of a smartphone
- 100 kJ : kinetic energy of a car at 50 km/h
- 1 MJ : running for 1 h
- 10 MJ : food energy of a person per day
- 200 MJ : daily use of a person (global average)

Industrial revolution

Global / industrial scale

- 1 MJ : 1 ton of water released from a 100 m dam
- 10 MJ : average solar radiation per m^2 per day in France
- 100 MJ : heat of combustion of 2 kg of oil
- 1 GJ : energy of an average gasoline reservoir (50 L)
- 1 TJ (10^{12} J) : consumption of a wide-body aircraft for a long-haul flight
- 1 PJ (10^{15} J) : 2 weeks of electricity production of a nuclear plant
- 1 EJ (10^{18} J) : 6 months of French electricity production
- 600 EJ : global annual consumption
- 50 000 EJ : proven reserves of fossil fuels
- 5 500 000 EJ : global solar radiation on Earth per year

1h of solar radiation on earth is a year of consumption

Plan of the lesson

1. What is energy?

2. The usable forms of energy

3. Exploitation of energy
   - Energy transformations
   - Exploitation by humans

Exploitation = transformation

- All progresses in harvesting energy aim at transforming one form of energy into another.

Water mill
Stream → Rotation



from wikimedia commons (Jean-Pol Grandmont, CC BY 2.0)

Steam engine
Thermal → Mechanical



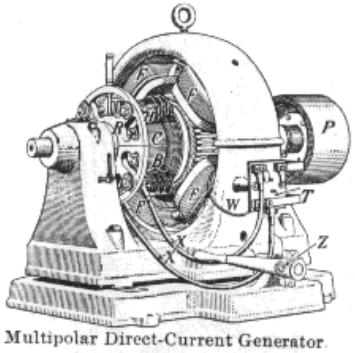
from wikimedia commons (André Karwath, CC BY-SA 2.5)



Battery
Chemical → Electrical

from wikimedia commons (Luigi Chiesa, CC BY 3.0)

Generator
Mechanical → Electrical



from wikimedia commons (public domain)

Control of fire
Chemical → thermal

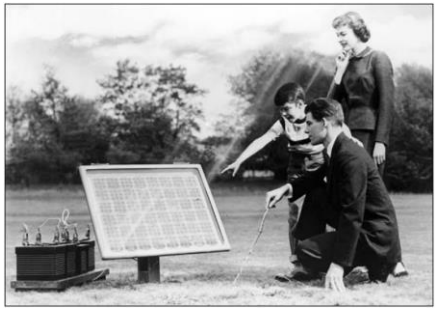




Light bulb
Electrical → Radiation

National Historical Park New Jersey

Photovoltaic
Radiation → Electrical



Look Magazine (1956)

List of energy transformations

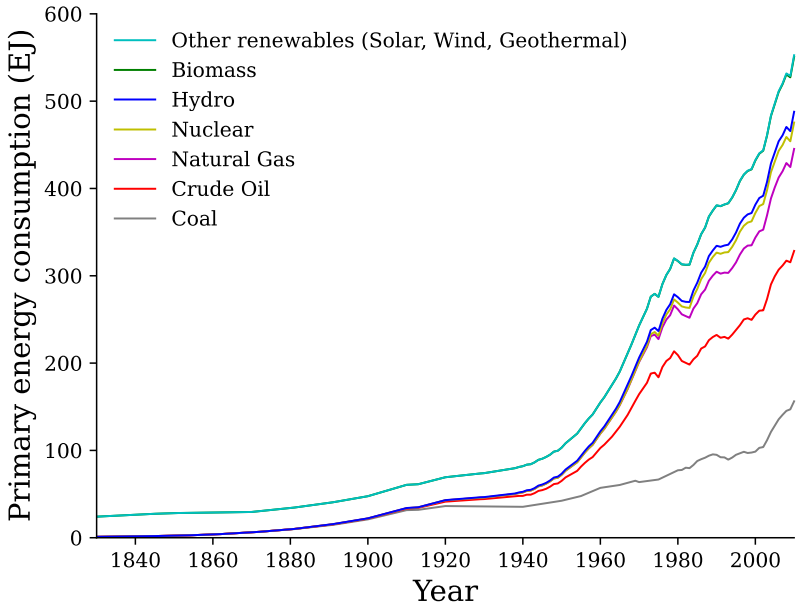
↓from \ to→	Mechanical	Thermal	Radiation	Electrical	Chemical	Nuclear
Mechanical	Gearbox	Friction	Synchrotron radiation	Generator	Metamorphism	Particle accelerator
Thermal	Turbine	Heat exchanger	Black body	Thermocouple (Seebeck effect)	Blast furnace	Supernova
Radiation	Radiometer	Thermal solar collector	Non linear optic	Photovoltaic cell	Photosynthesis	Photodesintegration
Electrical	Electrical engine	Electrical resistance	Lightning, LED	Transformator	Accumulator	-
Chemical	Muscle	Combustion	Luminescence	Fuel cell	Coal gasification	-
Nuclear	Atomic bomb	Nuclear reactor	Gamma radiation	Atomic battery	Radiolysis	Breeder reactor

- Efficiency of transformations can be limited by physics
  - Carnot limit (thermal energy)
  - Gibbs free enthalpy (electro-chemistry)
  - Shockley-Queisser limit (photovoltaic)
  - Betz limit (wind turbine)

Historical evolution and current use

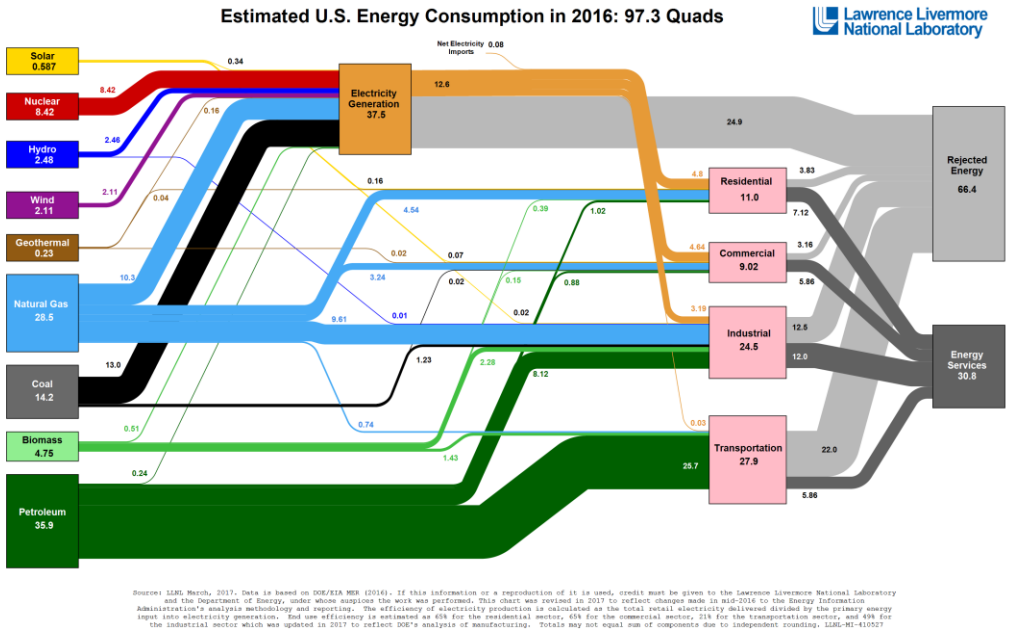
- Dominated by fossil fuels since the industrial revolution
- Usage ubiquitous in all economical sectors
- Final energy is only 1/3 of primary energy (efficiency, other losses)

Energy sources



From the Oil Drum

Energy use (US, 2014, in 10^{15} Btu = 1.055 EJ)



From Lawrence Livermore National Lab, Public Domain

Comparison between energy sources

- Fossil fuels are: abundant, highly concentrated

Type	Source	Resource
Renewable	Biofuels	1 TW
	Geothermal	1 TW
	Hydro	1-2 TW
	Wind	100 TW
	Solar	100 000 TW
Non renewable	Coal	20 000 EJ
	Oil	9 000 EJ
	Gas	36 000 EJ
	Nuclear	> 2 000 EJ

Power available (in W = J/s)

Energy reserve (proven)

global need : 600 EJ/yr ≈ 20 TW

Sources :
- Renewables: Jaffe & Taylor (2018) The Physics of Energy. Cambridge University Press
- Fossil fuels: British Petroleum
- Nuclear: IPCC

Electrical power produced per unit area of land

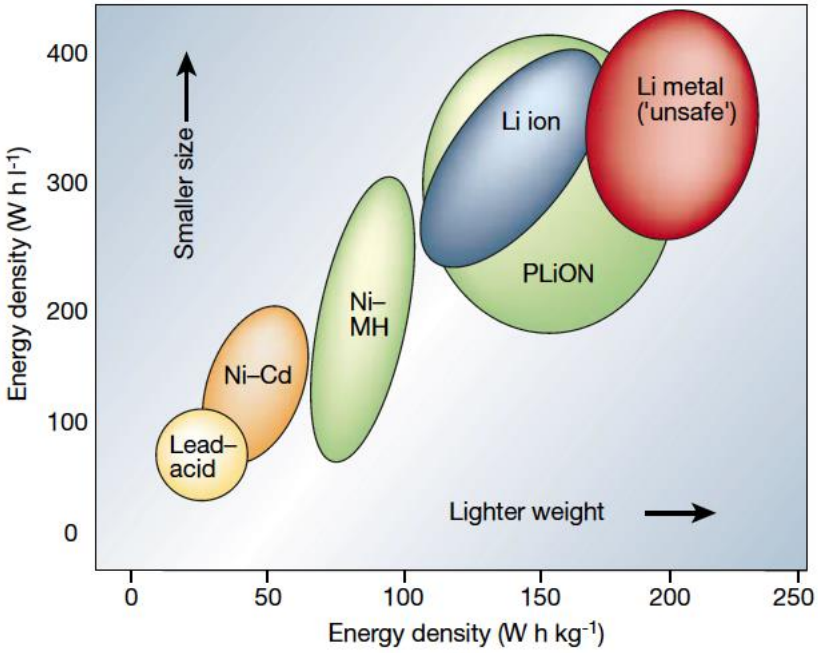
Type	Source	Power density
Renewable	Biofuels	0.23 W/m^2
Renewable	Hydro	11 W/m^2
Renewable	Solar	11 W/m^2
Non renewable	Coal (open cast)	22 W/m^2
Renewable	Geothermal	46 W/m^2
Renewable	Wind	114 W/m^2
Non renewable	Coal (underground)	570 W/m^2
Non renewable	Oil & gas	570 W/m^2
Non renewable	Nuclear	1 140 W/m^2

Source : Uwe R.Fritsche et al. (2017) Energy and Land Use - Global Land Outlook working paper. UNCCD and IRENA

Comparison between energy sources

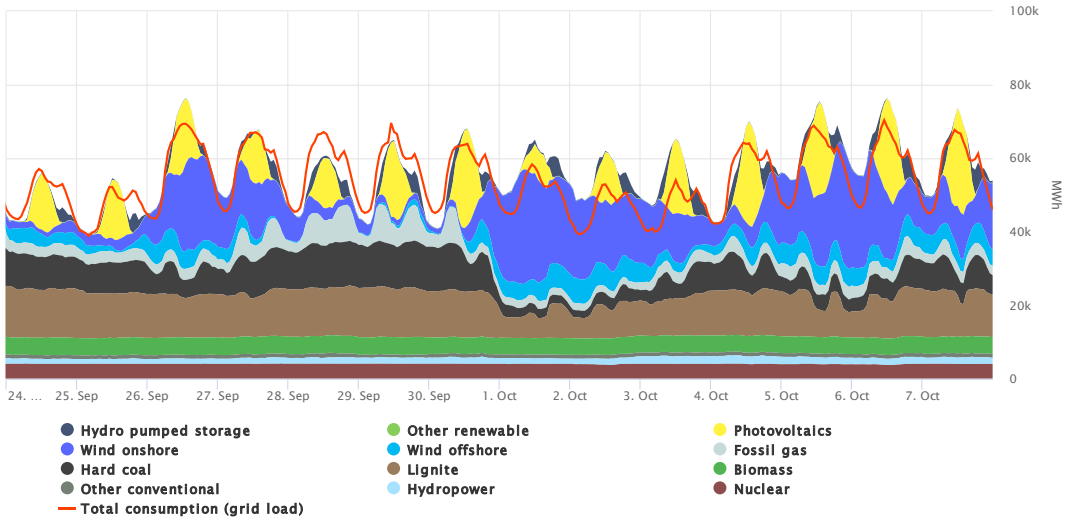
- Fossil fuels are: abundant, highly concentrated, easy to store and transform, on-demand, moderate investment / operation costs, energy but also used for matter…

Gasoline (with engine efficiency of 30%)
↑ 2750 W.h.L^{-1}



Tarascon & Armand (2001) Nature, 414(6861), 359-367

Electricity production in Germany (two weeks in 2022)

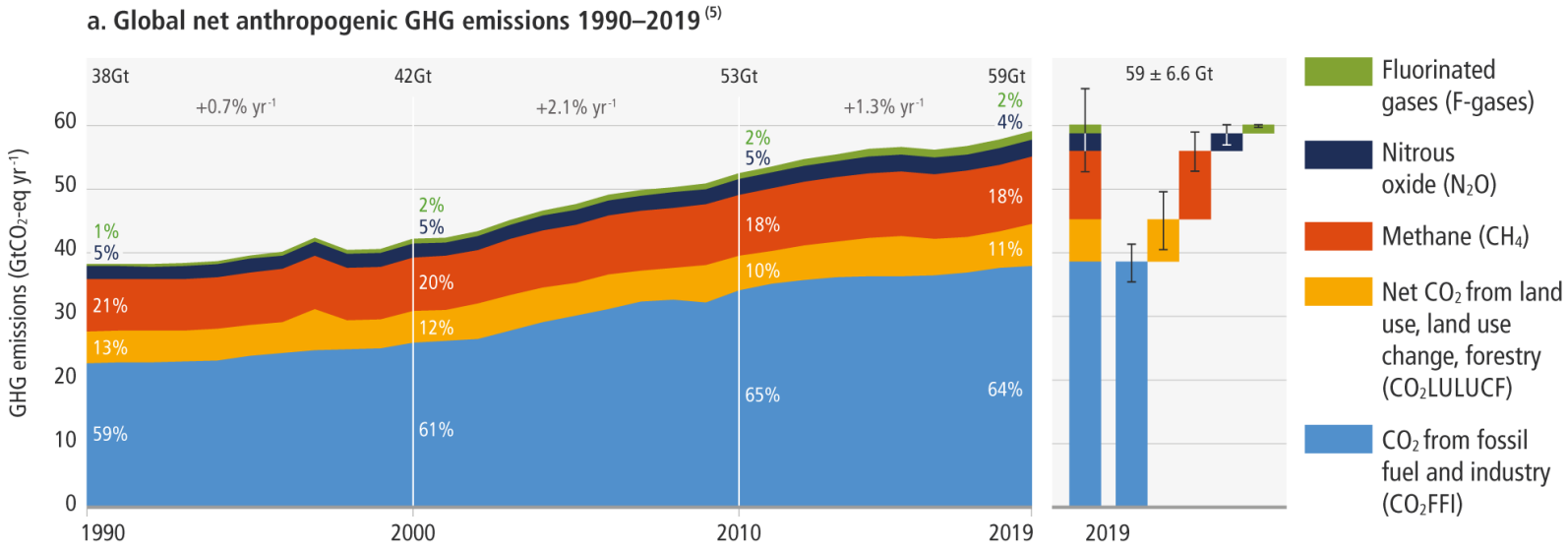


from wikimedia commons (by Bundesnetzagentur SMARD.de, CC BY-SA 4.0)

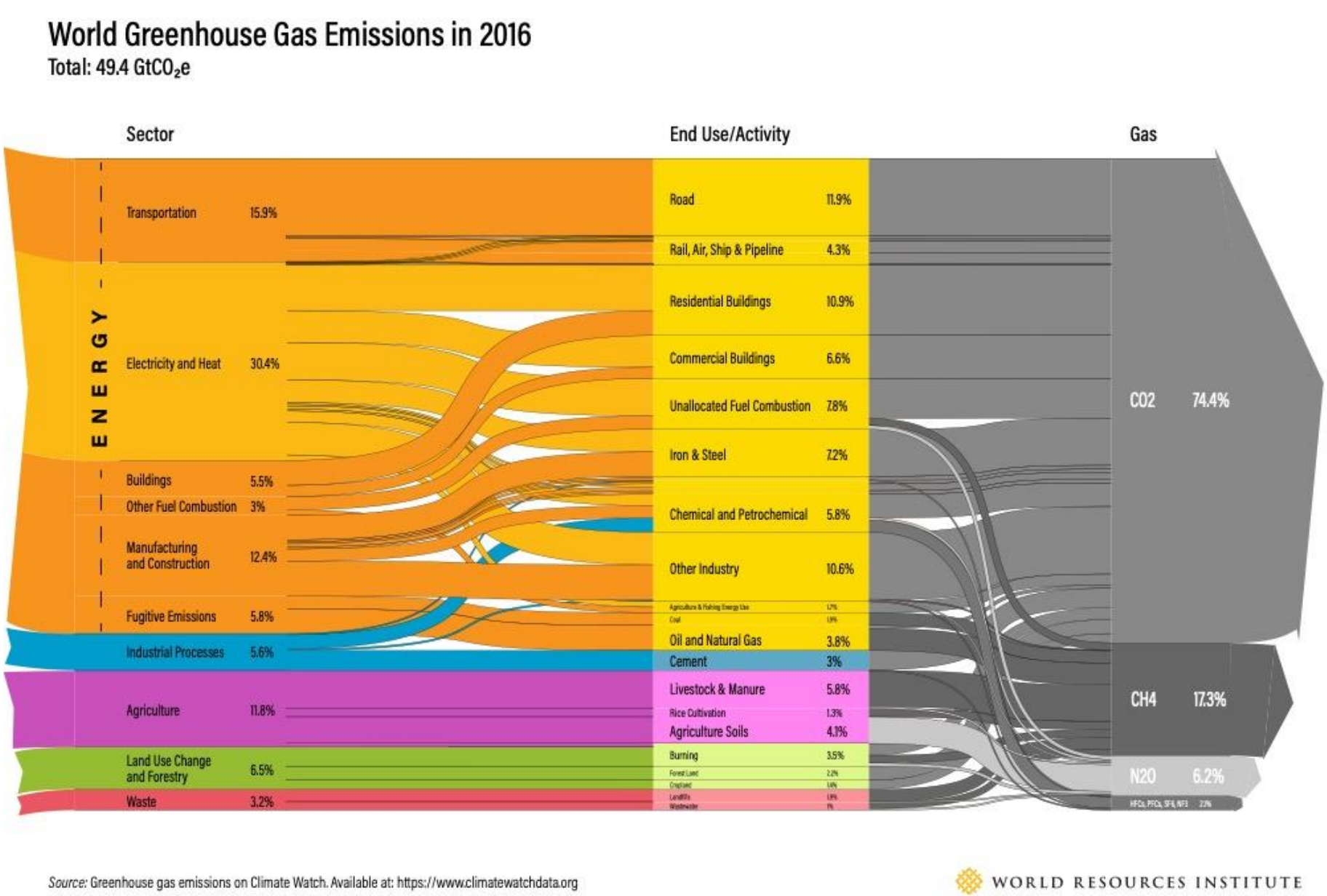
Gasoline
(with engine efficiency of 30%)
→ 3750 W.h.kg^{-1}

Externalities

- Fossil fuels are: abundant, highly concentrated, easy to store and transform, on-demand, moderate investment / operation costs, energy but also used for matter…
- But they are:
  - The main source of anthropogenic GHG emissions



Lamb, W. F., 2022, Data for Figure SPM.1 - Summary for Policymakers of the Working Group III Contribution to the IPCC Sixth Assessment Report



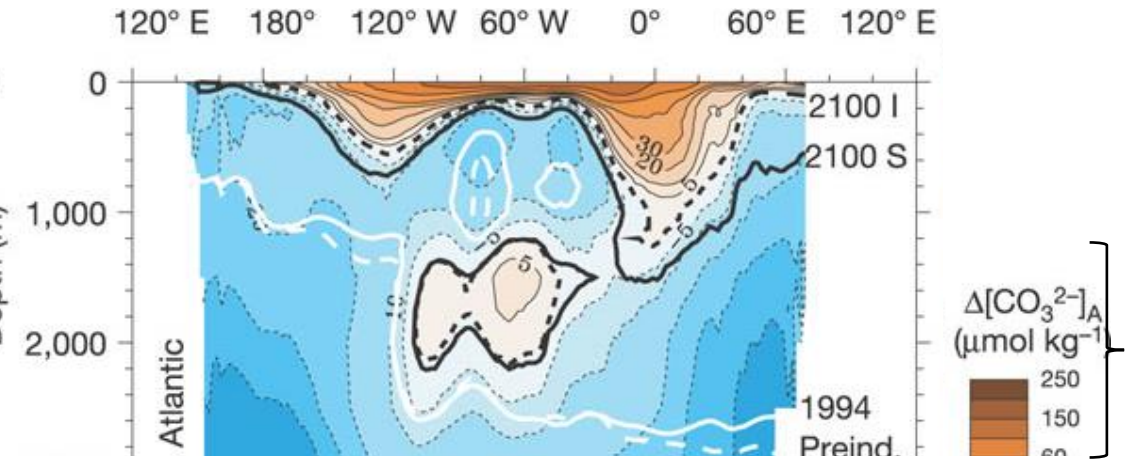
Source: Greenhouse gas emissions on Climate Watch. Available at: https://www.climatewatchdata.org

WORLD RESOURCES INSTITUTE

Externalities

- Fossil fuels are: abundant, highly concentrated, easy to store and transform, on-demand, moderate investment / operation costs…
- But they are:
  - The main source of anthropogenic GHG emissions
  - The main cause of ocean acidification

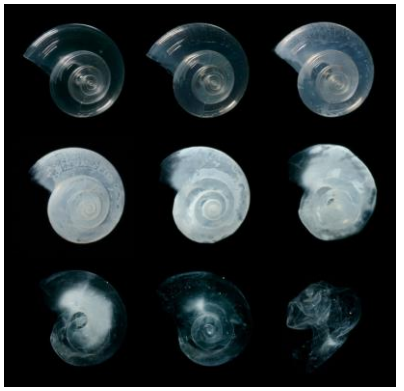
The aragonite saturation state in the year 2100



aragonite dissolves

Orr et al. (2005) Nature 437, 681-686 (2005)

Limacina helicina antarctica Pteropod shell exposed 45 days to the saturation state of aragonite in seawater as projected in the Southern Ocean by 2100



David Liittschwager/National Geographic

Externalities

- Fossil fuels are: abundant, highly concentrated, easy to store and transform, on-demand, moderate investment / operation costs…
- But they are:
  - The main source of anthropogenic GHG emissions
  - The main cause of ocean acidification
  - Major contributors to other pollutions : NOx, SOx, PM, VOCs…

Photochemical ozone pollution (Santiago, Chile)



from wikimedia commons (CC BY-SA 3.0)

air pollution ≈ 40 000 premature deaths per year in France (Santé Publique France)

Externalities

- Fossil fuels are: abundant, highly concentrated, easy to store and transform, on-demand, moderate investment / operation costs…
- But they are:
  - The main source of anthropogenic GHG emissions
  - The main cause of ocean acidification
  - Major contributors to other pollutions : NOx, SOx, PM, VOCs…
- Energy transition
  - Drastic change to sustainable energy (low carbon & long-term supply)
  - Paris agreement (COP21) : reach net zero carbon by mid-century

Key points

- **Noether's theorem**: energy as the conserved quantity associated with symmetry by time translation
- **Units of energy**: Joule as the SI unit (1 J = 1 $kg.m^2.s^{-2}$), alternatives units (eV, kcal, kW.h), equivalence with frequency and wavelength (spectroscopy) and with mass (relativity, nuclear processes)
- The different **forms of energy used by man** (elementary processes, scales of energy / radiation)
- Typical **orders of magnitude of energy use** (daily life, industrial use, global scale), and of energy reserves (fossil, solar)